

Coding & Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	Rising Waters – AI-Powered Flood Prediction and Early Warning System
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Dimple1603/HDI-Predictor https://github.com/Pallavi Sowreddi1603/Flood Risk-Predictor
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-Learn, Pandas, NumPy
Key Features Implemented	Flood Risk prediction, user input interface, data preprocessing, model integration, development category classification
Pending / Incomplete Features	Cloud deployment and advanced visual analytics
Setup / Run Instructions	Install dependencies using <code>pip install -r requirements.txt</code> and run using <code>python app.py</code>

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Partial
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project successfully implements a machine learning-based prediction system for estimating Human Development Index (HDI) values. The code follows a modular structure with separate components for preprocessing, model prediction, and web interface development. The application is easy to maintain and can be extended with additional machine learning algorithms and visualization features in future versions.